

Machine Learning Assignment

1. Machine Learning

Machine Learning is a branch of Artificial Intelligence that allows systems to learn from data and improve performance without being explicitly programmed. It focuses on building models that can make predictions or decisions.

2. Types of Machine Learning

Supervised Learning: Uses labeled data where input-output pairs are known. Examples include classification and regression problems.

Unsupervised Learning: Uses unlabeled data to find hidden patterns or groupings. Example: clustering.

Reinforcement Learning: Involves an agent learning through rewards and penalties by interacting with an environment.

3. Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary classification problems. It predicts the probability that a given input belongs to a particular class.

It uses a sigmoid function to map predictions between 0 and 1.

Formula: $P(Y=1) = 1 / (1 + e^{-z})$

Advantages: Simple, fast, interpretable.

Disadvantages: Limited to linear decision boundaries.

4. Implementation using Titanic Dataset

The Titanic dataset is used to predict whether a passenger survived or not based on features like age, gender, class, etc.

Steps Involved:

1. Import libraries 2. Load dataset 3. Data preprocessing 4. Train-test split 5. Model training 6. Prediction 7. Evaluation

Python Code

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

df = pd.read_csv("titanic.csv")

df = df.drop(['Name', 'Ticket', 'Cabin'], axis=1)

df['Age'].fillna(df['Age'].mean(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])
df['Embarked'] = le.fit_transform(df['Embarked'])

X = df.drop('Survived', axis=1)
y = df['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

5. Model Evaluation

Accuracy measures the proportion of correct predictions made by the model.

Confusion Matrix provides a detailed breakdown of predictions:

- True Positives: Correctly predicted survival
- True Negatives: Correctly predicted non-survival
- False Positives: Incorrect survival prediction
- False Negatives: Incorrect non-survival prediction

Conclusion

Logistic Regression is an effective and simple algorithm for binary classification tasks. In this assignment, we applied it to the Titanic dataset and evaluated its performance using accuracy and confusion matrix.